Pair of Straight Lines

Reference book: A text book on co-ordinate geometry by RAHMAN & BHATTACHARJEE

1. **Homogeneous equation (of degree n)**

An equation of the type

$$a_0y^n + a_1y^{n-1}\,x + a_2y^{n-2}\,x^2 + \cdots\cdots\cdots + a_ry^{n-r}\,x^r + \cdots\cdots + a_nx^n = 0$$

In which the sum of the powers of x and y in every term is the same (say n) is called a homogeneous equation (of degree n).

2. **Show that every homogeneous equation of the second degree represents a pair of straight lines passing through the origin.**

Solution: Let the general homogeneous equation of second degree is

$ax^2 + 2hxy + by^2 = 0$ (1)

Dividing both sides by x^2, we get

$a + 2h\left(\frac{y}{x}\right) + b\left(\frac{y}{x}\right)^2 = 0$

$\Rightarrow b\left(\frac{y}{x}\right)^2 + 2h\left(\frac{y}{x}\right) + a = 0$ (2)

Putting $\frac{y}{x} = p$

$\therefore bp^2 + 2hp + a = 0$ (3)

Which is a quadratic in p.

Let it roots be m_1 and m_2. Then equation (3) can be written as

$b(p - m_1)(p - m_2) = 0$

$\Rightarrow b\left(\frac{y}{x} - m_1\right)\left(\frac{y}{x} - m_2\right) = 0$

$\Rightarrow b(y - m_1x)(y - m_2x) = 0$

$\because$ if α and β are the roots of $ax^2 + bx + c = 0$, then we can write $ax^2 + bx + c = a(x - \alpha)(x - \beta)$

Thus equation (1) represents two straight lines

$y - m_1x = 0$

and

$y - m_2x = 0$

3. **Find the relation between the slopes of the two lines given by the homogeneous equation $ax^2 + 2hxy + by^2 = 0$.**

Solution: Let the general homogeneous equation of second degree is

$ax^2 + 2hxy + by^2 = 0$ (1)

Dividing both sides by x^2, we get

$a + 2h\left(\frac{y}{x}\right) + b\left(\frac{y}{x}\right)^2 = 0$

$\Rightarrow b\left(\frac{y}{x}\right)^2 + 2h\left(\frac{y}{x}\right) + a = 0$ (2)

Putting $\frac{y}{x} = p$

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$\Rightarrow b(y - m_1x)(y - m_2x) = 0$

$\because$ if α and β are the roots of $ax^2 + bx + c = 0$, then we can write $ax^2 + bx + c = a(x - \alpha)(x - \beta)$

Thus equation (1) represents two straight lines

$y - m_1x = 0$

and

$y - m_2x = 0$

Then the equation $ax^2 + 2hxy + by^2 = b(y - m_1x)(y - m_2x)$ must be identical.

$\Rightarrow b\left(\frac{a}{b}x^2 + \frac{2h}{b}xy + y^2\right) = b(y - m_1x)(y - m_2x)$

$\Rightarrow \left(y^2 + \frac{2h}{b}xy + \frac{a}{b}x\right) = (y - m_1x)(y - m_2x)$

$\Rightarrow y^2 + \frac{2h}{b}xy + \frac{a}{b}x = y^2 - m_2xy - m_1xy + m_1m_2x^2$

$\Rightarrow y^2 + \frac{2h}{b}xy + \frac{a}{b}x = y^2 - (m_1 + m_2)xy + m_1m_2x^2$

Equating coefficients of xy and x^2 we get

$\frac{2h}{b} = -(m_1 + m_2)$

$\Rightarrow (m_1 + m_2) = -\frac{2h}{b}$

and

$m_1 m_2 = \frac{a}{b}$

4. **Prove that a homogeneous equation of the n-th degree represents n straight lines, real or imaginary, which all pass through the origin.**

Solution: Consider the homogenous equation

$a_0 y^n + a_1 y^{n-1} x + a_2 y^{n-2} x^2 + \cdots\cdots\cdots + a_r y^{n-r} x^r + \cdots\cdots + a_n x^n = 0$…… (1)

This equation can be written as (dividing by x^n)

$a_0 \left(\frac{y}{x}\right)^n + a_1 \left(\frac{y}{x}\right)^{n-1} + a_2 \left(\frac{y}{x}\right)^{n-2} + \cdots\cdots\cdots + a_r \left(\frac{y}{x}\right)^{n-r} + \cdots\cdots + a_n = 0$ …….(2)

Since this is an equation of the n-th degree in $\frac{y}{x}$, it must have n roots. Let the roots of this equation be

$m_1, m_2, m_3, \ldots\ldots m_r, \ldots\ldots m_n$

Then the given equation (1) or (2) must be the same as

$\left(\frac{y}{x} - m_1\right)\left(\frac{y}{x} - m_2\right) \ldots\ldots \left(\frac{y}{x} - m_r\right) \ldots\ldots \left(\frac{y}{x} - m_n\right) = 0$

$\Rightarrow (y - m_1 x)(y - m_2 x) \ldots\ldots (y - m_r x) \ldots\ldots (y - m_n x) = 0$

This equation is satisfied by the points which satisfy the separate equations.

$(y - m_1 x) = 0, (y - m_2 x) = 0, \ldots\ldots (y - m_r x) = 0, \ldots\ldots (y - m_n x) = 0.$

which all pass through the origin. Conversely, the co-ordinates of all the points which satisfy these n equations satisfy equation (1) also.

5. **Find the angle between the straight lines $ax^2 + 2hxy + by^2 = 0$ (the axes are assumed to be rectangular).**

Solution: Let $ax^2 + 2hxy + by^2 = 0$ represent two straight lines

$y - m_1 x = 0$ …… (1)

and $y - m_2 x = 0$…… (2)

Then, $m_1 + m_2 = -\frac{2h}{b}$, $m_1 m_2 = \frac{a}{b}$

Let θ be the angle between the lines (1) and (2),

Then,

$\tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2}$

$$= \frac{\sqrt{(m_1 + m_2)^2 - 4m_1m_2}}{1 + m_1m_2}$$

$$= \frac{\sqrt{\frac{4h^2}{b^2} - 4\frac{a}{b}}}{1 + \frac{a}{b}}$$

$$\Rightarrow tan\theta = \frac{2\sqrt{h^2 - ab}}{a + b}$$

$$\therefore \theta = \tan^{-1}\frac{2\sqrt{h^2 - ab}}{a + b}$$

6. **Find the angle between the straight lines which represented by $ax^2 + 2hxy + by^2 = 0$ also find the condition of perpendicularity, parallelism, real and imaginary.**

Solution: Let $ax^2 + 2hxy + by^2 = 0$ represent two straight lines

$y - m_1x = 0$ …… (1)

and $y - m_2x = 0$…… (2)

Then, $m_1 + m_2 = -\frac{2h}{b}$, $m_1m_2 = \frac{a}{b}$

Let θ be the angle between the lines (1) and (2),

Then,

$$tan\theta = \frac{m_1 - m_2}{1 + m_1m_2}$$

$$= \frac{\sqrt{(m_1 + m_2)^2 - 4m_1m_2}}{1 + m_1m_2}$$

$$= \frac{\sqrt{\frac{4h^2}{b^2} - 4\frac{a}{b}}}{1 + \frac{a}{b}}$$

$$\Rightarrow tan\theta = \frac{2\sqrt{h^2 - ab}}{a + b}$$

$$\therefore \theta = \tan^{-1}\frac{2\sqrt{h^2 - ab}}{a + b} \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (iii)$$

Condition of perpendicularity:

If the straight lines perpendicularity to each other

then $\theta = 90^0$

$\therefore tan\theta = \infty$

Now from (iii), we get

$a + b = 0$

Condition of parallelism or Coincidence:

If the straight lines be coincident

then $\theta = 0^0$

$\therefore tan\theta = 0$

Now from (iii), we get

$ab = h^2$

Condition of real:

Two lines represented by $ax^2 + 2hxy + by^2 = 0$ will be real if $h^2 > ab$.

Condition of imaginary:

Two lines represented by $ax^2 + 2hxy + by^2 = 0$ will be imaginary if $h^2 < ab$.

7. **Find the equation of bisectors of the angles between the pair of lines represented by $\boldsymbol{ax^2 + 2hxy + by^2 = 0}$**

Solution: Let $ax^2 + 2hxy + by^2 = 0$ represent two straight lines

$y - m_1x = 0$ ..... (1)

and $y - m_2x = 0$..... (2)

Then, $m_1 + m_2 = -\frac{2h}{b}$, $m_1m_2 = \frac{a}{b}$

Now, the equation of bisectors of the angles between the lines (1) and (2) is given by,

$$\frac{y - m_1x}{\sqrt{1 + {m_1}^2}} = \pm\frac{y - m_2x}{\sqrt{1 + {m_2}^2}}$$

$\Rightarrow (y - m_1x)^2(1 + {m_2}^2) = (y - m_2x)^2(1 + {m_1}^2)$

$\Rightarrow (y^2 - 2m_1xy + {m_1}^2x^2)(1 + {m_2}^2) = (y^2 - 2m_2xy + {m_2}^2x^2)(1 + {m_1}^2)$

$\Rightarrow y^2 - 2m_1xy + {m_1}^2x^2 + {m_2}^2y^2 - 2m_1{m_2}^2xy + {m_1}^2{m_2}^2x^2$
$= y^2 - 2m_2xy + {m_2}^2x^2 + {m_1}^2y^2 - 2m_2{m_1}^2xy + {m_1}^2{m_2}^2x^2$

$\Rightarrow -2m_1xy + {m_1}^2x^2 + {m_2}^2y^2 - 2m_1{m_2}^2xy + 2m_2xy - {m_2}^2x^2 - {m_1}^2y^2$
$+ 2m_2{m_1}^2xy = 0$

$\Rightarrow -2(m_1 - m_2)xy + ({m_1}^2 - {m_2}^2)x^2 - ({m_1}^2 - {m_2}^2)y^2 + 2m_1m_2(m_1 - m_2)xy = 0$

$\Rightarrow -2xy + (m_1 + m_2)x^2 - (m_1 + m_2)y^2 + 2m_1m_2xy = 0$

$\Rightarrow -2xy - \frac{2h}{b}x^2 + \frac{2h}{b}y^2 + 2\frac{a}{b}xy = 0$

$\Rightarrow -bxy - hx^2 + hy^2 + axy = 0$

$\Rightarrow h(x^2 - y^2) = (a - b)xy$

$\therefore \frac{x^2 - y^2}{a - b} = \frac{xy}{h}$

8. **Prove that the product of the perpendiculars from (x_1, y_1) on the lines given by $ax^2 + 2hxy + by^2 = 0$ is $\frac{a{x_1}^2 + 2hx_1y_1 + b{y_1}^2}{\sqrt{(a-b)^2 + 4h^2}}$**

Solution: Let $ax^2 + 2hxy + by^2 = 0$ represent two straight lines,

$y - m_1x = 0$ ..... (1)

And $y - m_2x = 0$..... (2)

Then, $m_1 + m_2 = -\frac{2h}{b}$, $m_1m_2 = \frac{a}{b}$

Let d_1 be the perpendicular distance from (x_1, y_1)on (1), then,

$$d_1 = \frac{y_1 - m_1x_1}{\sqrt{1 + m_1^2}}$$

Again, let d_2 be the perpendicular distance from (x_1, y_1) on (2),then,

$$d_2 = \frac{y_1 - m_2x_1}{\sqrt{1 + m_2^2}}$$

Now the product of the perpendicular

$$= \frac{(y_1 - m_1x_1)}{\sqrt{1 + m_1{}^2}} \frac{(y_1 - m_2x_1)}{\sqrt{1 + m_2{}^2}}$$

$$= \frac{y_1{}^2 - (m_1 + m_2)x_1y_1 + m_1m_2x_1{}^2}{\sqrt{1 + m_1{}^2 + m_2{}^2 + m_1{}^2m_2{}^2}}$$

$$= \frac{y_1{}^2 + \frac{2h}{b}x_1y_1 + \frac{a}{b}x_1{}^2}{\sqrt{1 + (m_1 + m_2)^2 - 2m_1m_2 + m_1{}^2m_2{}^2}}$$

$$= \frac{y_1{}^2 + \frac{2h}{b}x_1y_1 + \frac{a}{b}x_1{}^2}{\sqrt{1 + \frac{4h^2}{b^2} - 2\frac{a}{b} + \frac{a^2}{b^2}}}$$

$$= \frac{by_1{}^2 + 2hx_1y_1 + ax_1{}^2}{\sqrt{b^2 + 4h^2 - 2ab + a^2}}$$

$$= \frac{ax_1{}^2 + 2hx_1y_1 + by_1{}^2}{\sqrt{(a-b)^2 + 4h^2}}$$ (Proved).

9. **If two straight lines represented by the equation $x^2(tan^2\theta + cos^2\theta) - 2xytan\theta + y^2sin^2\theta = 0$ makes an angle α and β with x −axis respectively, then show that $tan\alpha - tan\beta = 2$.**

Solution:

Given equation $x^2(tan^2\theta + cos^2\theta) - 2xytan\theta + y^2sin^2\theta = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$

Since the two lines represented by the equation(i) makes an angle α and β with x −axis.

So their equation are

$y - xtan\alpha = 0$ and $y - xtan\beta = 0$

$$\therefore tan\alpha + tan\beta = \frac{2tan\theta}{sin^2\theta}$$

$$= 2\frac{sin\theta}{cos\theta}.\frac{1}{sin^2\theta}$$

$$= 2sec\theta\ cosec\theta$$

and $tan\alpha.tan\beta = \frac{tan^2\theta+cos^2\theta}{sin^2\theta}$

$$= \frac{sin^2\theta}{cos^2\theta}.\frac{1}{sin^2\theta} + \frac{cos^2\theta}{sin^2\theta}$$

$$= sec^2\theta + cot^2\theta$$

Now $(tan\alpha - tan\beta)^2 = (tan\alpha + tan\beta)^2 - 4tan\alpha.tan\beta$

$$= (2sec\theta\ cosec\theta)^2 - 4(sec^2\theta + cot^2\theta)$$
$$= 4sec^2\theta\ cosec^2\theta - 4sec^2\theta - 4cot^2\theta$$
$$= 4\{sec^2\theta(cosec^2\theta - 1)\} - 4cot^2\theta$$
$$= 4sec^2\theta\ cot^2\theta - 4cot^2\theta$$
$$= 4cot^2\theta(sec^2\theta - 1)$$
$$= 4cot^2\theta\ tan^2\theta$$

$(tan\alpha - tan\beta)^2 = 4$

$\therefore\ tan\alpha - tan\beta = 2$

10. **The condition that the pair of lines $ax^2+2hxy+by^2=0$ and $a'x^2+2h'xy+b'y^2=0$ Should have one line in common is $4(ah' - a'h)(hb' - h'b) = (ab' - a'b)^2$.**

Solution: Given equations

$ax^2+2hxy+by^2=0$…….. (1)

and $a'x^2+2h'xy+b'y^2=0$…… (2)

let $y = mx$ be the common line of (1) and (2), then $y = mx$ satisfied both (1) and (2), we have

$ax^2+2hx.mx+b.m^2x^2=0$

$\Rightarrow bm^2 + 2hm + a=0$ …….. (3)

And $a'x^2+2h'x.mx+b'.m^2x^2.=0$

$\Rightarrow b'm^2+2h'm+a'=0$ …….. (4)

Now from (3) and (4), we have,

$$\frac{m^2}{2ah'-2a'h}=\frac{m}{ab'-a'b}=\frac{1}{2hb'-2h'b}$$

$$\Rightarrow m=\frac{2(ah'-a'h)}{ab'-a'b}, \qquad m=\frac{ab'-a'b}{2(hb'-h'b)}$$

$$\Rightarrow \frac{2(ah'-a'h)}{ab'-a'b}=\frac{ab'-a'b}{2(hb'-h'b)}$$

$\Rightarrow 4(ah'-a'h)(hb'-h'b)=(ab'-a'b)^2$ (Proved)

11. **Show that the line $Ax+By+C=0$ and the two lines represented by $(Ax+By)^2-3(Ay-Bx)^2=0$ from the sides of an equilateral triangle.**

Solution: Given that,

$Ax+By+C=0$ … … … . . (1)

And

$(Ax+By)^2=3(Ay-Bx)^2$

$\therefore\ Ax+By=\pm\sqrt{3}(Ay-Bx)$

$\Rightarrow \left(A+\sqrt{3}\,B\right)x+\left(B-\sqrt{3}A\right)y=0$ … … … … (2)

And

$\Rightarrow \left(A-\sqrt{3}\,B\right)x+\left(B+\sqrt{3}A\right)y=0$ … … … … (3)

Now, the slopes of (1), (2) & (3) are

$$-A/_B, \qquad -\frac{A+\sqrt{3}\,B}{B-\sqrt{3}\,A} \quad and \quad -\frac{A-\sqrt{3}\,B}{B+\sqrt{3}\,A}\ repectively.$$

Let θ be the angle between (1) & (2), then

$$\tan\theta=\frac{-A/_B+\frac{A+\sqrt{3}\,B}{B-\sqrt{3}\,A}}{1+\left(-A/_B\right)\left(-\frac{A+\sqrt{3}\,B}{B-\sqrt{3}\,A}\right)}$$

$$=\frac{-AB+\sqrt{3}A^2+AB+\sqrt{3}B^2}{B^2-\sqrt{3}AB+A^2+\sqrt{3}AB}$$

$$= \frac{\sqrt{3}(A^2 + B^2)}{A^2 + B^2}$$

$$= \sqrt{3} = \tan 60°$$

$\therefore\ \theta = 60°$

Again, let φ be the angle between (2) and (3), then

$$\tan\varphi = \frac{-\frac{A+\sqrt{3}\,B}{B-\sqrt{3}\,A} + \frac{A-\sqrt{3}\,B}{B+\sqrt{3}\,A}}{1 + \left(-\frac{A+\sqrt{3}\,B}{B-\sqrt{3}\,A}\right)\left(-\frac{A-\sqrt{3}\,B}{B+\sqrt{3}\,A}\right)} = \sqrt{3} = \tan 60°$$

$\therefore\ \varphi = 60°$

Therefore, the two angles are equal.

Hence the given equations form an equilateral triangle.

12. **Find the area of the triangle formed by the lines $ax^2 + 2hxy + by^2 = 0$ and $lx + my + n = 0$.**

Solution: Given equations

$ax^2 + 2hxy + by^2 = 0$ ... ... (1)

and $lx + my + n = 0$ ... ... (2)

let the equation (1) represent two straight lines

y$-m_1x = 0$..... (3)

and y$-m_2x = 0$..... (4)

Then $m_1 + m_2 = -\frac{2h}{b}$ and $m_1m_2 = \frac{a}{b}$

From (3) and (4), it is clear that one of the vertices of the triangle is at (0, 0).

Solving (2) and (3), we get

$lx + mm_1x + n = 0$

$\Rightarrow\ x = -\frac{n}{l + mm_1}$

$\therefore\ y = -\frac{m_1n}{l + mm_1}$

Thus we have a vertex as A $\left(-\frac{n}{l+mm_1},\ -\frac{m_1n}{l+mm_1}\right)$

Similarly, solving (2) and (4), we get another vertex as B $\left(-\frac{n}{l+mm_2},\ -\frac{m_2n}{l+mm_2}\right)$

Hence the area of the triangle is

$$\frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ -\frac{n}{l+mm_1} & -\frac{m_1 n}{l+mm_1} & 1 \\ -\frac{n}{l+mm_2} & -\frac{m_2 n}{l+mm_2} & 1 \end{vmatrix}$$

$$= \frac{1}{2}\left[\frac{m_2 n^2}{(l+mm_1)(l+mm_2)} - \frac{m_1 n^2}{(l+mm_1)(l+mm_2)}\right]$$

$$= \frac{n^2}{2}\left[\frac{m_2 - m_1}{(l+mm_1)(l+mm_2)}\right]$$

$$= \frac{n^2}{2}\left[\frac{\sqrt{(m_1+m_2)^2 - 4m_1 m_2}}{l^2 + lm(m_1+m_2) + m^2 m_1 m_2}\right]$$

$$= \frac{n^2}{2}\left[\frac{\sqrt{\frac{4h^2}{b^2} - \frac{4a}{b}}}{l^2 - \frac{2hlm}{b} + m^2.\frac{a}{b}}\right]$$

$$= \frac{n^2}{2}\left[\frac{\sqrt{4h^2 - 4ab}}{bl^2 - 2hlm + am^2}\right]$$

$$= n^2 \frac{\sqrt{h^2 - ab}}{bl^2 - 2hlm + am^2} \qquad \text{(Proved)}$$

13. Find the condition that the general equation of 2nd degree should represent a pair of straight lines.

Solution: Let, $l_1x + m_1y + n_1 = 0$ And $l_2x + m_2y + n_2 = 0$ be two straight lines represented by

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

We can write,

$$(l_1x + m_1y + n_1)(l_2x + m_2y + n_2) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c$$

$$l_1l_2x^2 + l_1m_2xy + l_1n_2x + m_1l_2xy + m_1m_2y^2 + m_1n_2y + n_1l_2x + n_1m_2y + n_1n_2 = ax^2 + 2hxy + by^2 + 2gx + 2fy + c$$

$$l_1l_2x^2 + (l_1m_2 + l_2m_1)xy + m_1m_2y^2 + (l_1n_2 + l_2n_1)x + (m_1n_2 + m_2n_1)y + n_1n_2 = ax^2 + 2hxy + by^2 + 2gx + 2fy + c$$

Equating the co-efficient like term from both sides,

$$l_1l_2 = a \ldots\ldots\ldots\ldots (1)$$
$$m_1m_2 = b \ldots\ldots\ldots\ldots (2)$$
$$n_1n_2 = c \ldots\ldots\ldots\ldots (3)$$
$$l_1m_2 + l_2m_1 = 2h \ldots\ldots\ldots\ldots (4)$$
$$l_1n_2 + l_2n_1 = 2g \ldots\ldots\ldots\ldots\ldots (5)$$
$$m_1n_2 + m_2n_1 = 2f \ldots\ldots\ldots\ldots\ldots (6)$$

From (4), (5) & (6), we get

$$8fgh = (l_1m_2 + l_2m_1)(l_1n_2 + l_2n_1)(m_1n_2 + m_2n_1)$$

$$= l_1l_2(m_1{}^2n_2{}^2 + m_2{}^2n_1{}^2) + m_1m_2\left(l_1{}^2n_2{}^2 + l_2{}^2n_1{}^2\right) + n_1n_2\left(l_1{}^2m_2{}^2 + l_2{}^2m_1{}^2\right) + 2l_1l_2m_1m_2n_1n_2$$

$$= a\{(m_1n_2 + m_2n_1)^2 - 2m_1m_2n_1n_2\} + b\{(l_1n_2 + l_2n_1)^2 - 2l_1l_2n_1n_2\} + c\{(l_1m_2 + l_2m_1)^2 - 2l_1l_2m_1m_2\} + 2l_1l_2m_1m_2n_1n_2$$

$$= a\{4f^2 - 2bc\} + b\{4g^2 - 2ca\} + c\{4h^2 - 2ab\} + 2abc$$

$$\therefore\ 2fgh = af^2 + bg^2 + ch^2 - abc$$

$$\Rightarrow\ abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

That is, $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$

$\therefore\ \Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

which is the required condition.

14. **The lines represented by $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ the equidistant from the origin, then prove that $f^4 - g^4 = c(bf^2 - ag^2)$ and $h(g^2 - f^2) = fg(a - b)$.**

Solution: let the equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \ldots\ldots\ldots\ldots (1)$$

Represent two straight lines

$$l_1x + m_1y + n_1 = 0 \ldots\ldots\ldots\ldots\ldots (2)$$

and $\quad l_2x + m_2y + n_2 = 0 \ldots\ldots\ldots\ldots\ldots (3)$

Then $(l_1x + m_1y + n_1)(l_2x + m_2y + n_2) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c$

Equating the co-efficient of like powers terms from both sides, we get

$$l_1l_2 = a, m_1m_2 = b, n_1n_2 = c, l_1m_2 + l_2m_1 = 2h, l_1n_2 + l_2n_1 = 2g, m_1n_2 + m_2n_1 = 2f$$

According to the condition,

$$\frac{n_1}{\sqrt{{l_1}^2 + {m_1}^2}} = \pm \frac{n_2}{\sqrt{{l_2}^2 + {m_2}^2}}$$

$$\Rightarrow {n_1}^2({l_2}^2 + {m_2}^2) = {n_2}^2({l_1}^2 + {m_1}^2)$$

$$\Rightarrow {n_1}^2{l_2}^2 + {n_1}^2{m_2}^2 = {n_2}^2{l_1}^2 + {n_2}^2{m_1}^2$$

$$\Rightarrow {n_1}^2{l_2}^2 - {n_2}^2{l_1}^2 = {n_2}^2{m_1}^2 - {n_1}^2{m_2}^2$$

$$\Rightarrow (l_1n_2 + l_2n_1)(l_1n_2 - l_2n_1) = (m_2n_1 + m_1n_2)(m_2n_1 - m_1n_2)$$

$$\Rightarrow (l_1n_2 + l_2n_1)\sqrt{(l_1n_2 + l_2n_1)^2 - 4l_1l_2n_1n_2} = (m_2n_1 + m_1n_2)\sqrt{(m_2n_1 + m_1n_2)^2 - 4m_1m_2n_1n_2}$$

$$\Rightarrow 2g\sqrt{4g^2 - 4ac} = 2f\sqrt{4f^2 - 4bc}$$

$$\Rightarrow g\sqrt{g^2 - ac} = f\sqrt{f^2 - bc}$$

$$\therefore\ f^4 - g^4 = c(bf^2 - ag^2)$$

Second part:

$$f^4 - g^4 = c(bf^2 - ag^2)$$

Since the equation (1) represents the pair of straight lines, then

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$\Rightarrow c(ab - h^2) = af^2 + bg^2 - 2fgh$$

$$\therefore\ c = \frac{af^2 + bg^2 - 2fgh}{ab - h^2}$$

$$\therefore\ (f^4 - g^4)(ab - h^2) = (af^2 + bg^2 - 2fgh)(bf^2 - ag^2)$$

$$\Rightarrow abg^4 - g^4h^2 - abf^4 + f^4h^2 = a^2f^2g^2 + abg^4 - 2afg^3h - abf^4 - b^2f^2g^2 + 2bfg^3h$$

$$\Rightarrow g^2(a^2f^2 - 2afgh - g^2h^2) = f^2(b^2g^2 - 2bfgh + f^2h^2)$$

$$\Rightarrow g^2(af - gh)^2 = f^2(bg - fh)^2$$

$$\Rightarrow g(af - gh) = f(bg - fh)$$

$$\Rightarrow h(g^2 - f^2) = fg(a - b) \qquad (Proved)$$

15. **If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represent two parallel straight lines , then show that $a/h = h/b = g/f$ and find the distance between them.**

Solution: since the equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \ldots\ldots\ldots\ldots\ldots (1)$$

Represent two parallel straight lines, then we get

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0 \ldots\ldots\ldots\ldots (2)$$

$$And \qquad ab = h^2 \ldots\ldots\ldots\ldots\ldots\ldots (3)$$

From (3), we get

$$a/h = h/b$$

From (2), we get

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$
$$\Rightarrow c(ab - h^2) + 2fgh - af^2 - bg^2 = 0$$
$$\Rightarrow af^2 - 2fgh + bg^2 = 0$$
$$\Rightarrow af^2 - 2fg\sqrt{ab} + bg^2 = 0$$
$$\Rightarrow (\sqrt{a}\, f - \sqrt{b}\, g)^2 = 0$$
$$\Rightarrow \sqrt{a}\, f - \sqrt{b}\, g = 0$$
$$\Rightarrow \sqrt{a}\, f = \sqrt{b}\, g$$
$$\Rightarrow \frac{\sqrt{a}}{\sqrt{b}} = \frac{g}{f}$$
$$\Rightarrow \frac{a}{\sqrt{ab}} = \frac{a}{h} = \frac{g}{f}$$
$$\Rightarrow \frac{a}{h} = \frac{g}{f}$$
$$\Rightarrow \frac{a}{h} = \frac{h}{b} = \frac{g}{f}$$

(Proved).

Second part: let the equation (1) represent two parallel straight lines

$$lx + my + n_1 = 0$$
$$and \qquad lx + my + n_2 = 0$$

Then the distance between them is given by

$$P = \left|\frac{n_2 - n_1}{\sqrt{l^2 + m^2}}\right|$$

Now,

$$(lx + my + n_1)(lx + my + n_2) = \ ax^2 + 2hxy + by^2 + 2gx + 2fy + c$$

Equating the co-efficient of terms of line powers,

$$l^2 = a, m^2 \ = b, n_1 n_2 = c, 2lm = 2h, l(n_1 + n_2) = 2g, m(n_1 + n_2) = 2f$$

$$\therefore P = \frac{\sqrt{(n_1 + n_2)^2 - 4n_1 n_2}}{\sqrt{l^2 + m^2}} = \frac{\sqrt{\frac{4g^2}{l^2} - 4c}}{\sqrt{a + b}} = 2\frac{\sqrt{g^2 - 4ac}}{\sqrt{a(a + b)}} \qquad \text{(Ans)}$$

16. **If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents two straight lines, prove that the square of the distance of their point of intersection from the origin is $\frac{c(a+b) - f^2 - g^2}{ab - h^2}$**

Solution: let the equation

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \ldots\ldots\ldots\ldots\ldots (1)$$

Represent two straight lines

$$l_1 x + m_1 y + n_1 = 0 \ldots\ldots\ldots\ldots\ldots (2)$$

and $$l_2 x + m_2 y + n_2 = 0 \ldots\ldots\ldots\ldots\ldots (3)$$

Then $(l_1 x + m_1 y + n_1)(l_2 x + m_2 y + n_2) = \ ax^2 + 2hxy + by^2 + 2gx + 2fy + c$

Equating the co-efficient of like powers terms from both sides, we get

$$l_1 l_2 = a, m_1 m_2 = b, n_1 n_2 = c, l_1 m_2 + l_2 m_1 = 2h, l_1 n_2 + l_2 n_1 = 2g, m_1 n_2 + m_2 n_1 = 2f$$

Now from (2) and (3), we get

$$\frac{x}{m_1 n_2 - m_2 n_1} = \frac{y}{n_1 l_2 - n_2 l_1} = \frac{1}{l_1 m_2 - l_2 m_1}$$

$$\Rightarrow x = \frac{m_1 n_2 - m_2 n_1}{l_1 m_2 - l_2 m_1}, \ \ y = \frac{n_1 l_2 - n_2 l_1}{l_1 m_2 - l_2 m_1}$$

So the square of the distance of their point of intersection from the origin is

$$= \left(\sqrt{(x - 0)^2 + (y - 0)^2}\right)^2$$

$$= x^2 + y^2$$

$$= \left(\frac{m_1 n_2 - m_2 n_1}{l_1 m_2 - l_2 m_1}\right)^2 + \left(\frac{n_1 l_2 - n_2 l_1}{l_1 m_2 - l_2 m_1}\right)^2$$

$$= \frac{(m_1n_2 - m_2n_1)^2 + (n_1l_2 - n_2l_1)^2}{(l_1m_2 - l_2m_1)^2}$$

$$= \frac{(m_1n_2 + m_2n_1)^2 - 4m_1m_2n_1n_2 + (n_1l_2 + n_2l_1)^2 - 4n_1n_2l_1l_2}{(l_1m_2 + l_2m_1)^2 - 4\,l_1l_2m_1m_2}$$

$$= \frac{4f^2 - 4bc + 4g^2 - 4ca}{4h^2 - 4ab}$$

$$= \frac{c(a+b) - f^2 - g^2}{ab - h^2}$$

17. **Find the angle between the pair of straight lines which represented by $3x^2 - 5xy - 2y^2 = 0$.**

Solution:

Consider the equation

$ax^2 + 2hxy + by^2 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$

Given equation is

$3x^2 - 5xy - 2y^2 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$

Comparing equation (i) and (ii), we get

$a = 3, h = -\frac{5}{2}, b = -2$

Let θ be the angle

$$\theta = \tan^{-1}\frac{2\sqrt{h^2 - ab}}{a + b}$$

$$\Rightarrow \theta = \tan^{-1}\frac{2\sqrt{\left(-\frac{5}{2}\right)^2 - 3(-2)}}{3 - 2}$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{2\sqrt{\frac{25}{4}+6}}{1}\right)$$

$$\Rightarrow \theta = tan^{-1}\left(2\sqrt{\frac{25}{4}+6}\right)$$

$$\Rightarrow \theta = tan^{-1}\left(2\sqrt{\frac{25+24}{4}}\right)$$

$$\Rightarrow \theta = tan^{-1}\left(2\sqrt{\frac{49}{4}}\right)$$

$$\Rightarrow \theta = tan^{-1}\left(2.\frac{7}{2}\right)$$

$$\Rightarrow \theta = tan^{-1}(7)$$

18. **Prove that the equation $12x^2 - 10xy + 2y^2 + 11x - 5y + 2 = 0$ represents pair of straight lines.**

Solution:

General equation of second degree is

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$

Given equation is

$12x^2 - 10xy + 2y^2 + 11x - 5y + 2 = 0 \ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$

Comparing equation (i) and (ii), we get

$a = 12, h = -5, b = 2, g = \frac{11}{2}, f = -\frac{5}{2}, c = 2$

$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$

$$\Rightarrow \Delta = 12.2.2 + 2.-\frac{5}{2}.\frac{11}{2}.(-5) - 12.\left(-\frac{5}{2}\right)^2 - 2.\left(\frac{11}{2}\right)^2 - 2(-5)^2$$

$$\Rightarrow \Delta = 48 + \frac{25.11}{2} - 12.\frac{25}{4} - 2.\frac{121}{4} - 2.25$$

$$\Rightarrow \Delta = 48 + \frac{275}{2} - 75 - \frac{121}{2} - 50$$

$$\Rightarrow \Delta = -77 + \frac{275}{2} - \frac{121}{2}$$

$$\Rightarrow \Delta = \frac{-154 + 275 - 121}{2}$$

$$\Rightarrow \Delta = \frac{-275 + 275}{2}$$

$$\Rightarrow \Delta = 0$$

So this given equation can be represents as a pair of straight lines.

19. **Find the value of k so that the equation $kx^2 - 10xy + 2y^2 + 11x - 5y + 2 = 0$ may represent pair of straight lines.**

Solution:

General equation of second degree is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$$

Given equation is

$$kx^2 - 10xy + 2y^2 + 11x - 5y + 2 = 0 \ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$$

Comparing equation (i) and (ii), we get

$$a = k, h = -5, b = 2, g = \tfrac{11}{2}, f = -\tfrac{5}{2}, c = 2$$

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

$$\Rightarrow k.2.2 + 2.-\frac{5}{2}.\frac{11}{2}.(-5) - k.\left(-\frac{5}{2}\right)^2 - 2.\left(\frac{11}{2}\right)^2 - 2(-5)^2 = 0$$

$$\Rightarrow 4k + \frac{25.11}{2} - k.\frac{25}{4} - 2.\frac{121}{4} - 2.25 = 0$$

$$\Rightarrow 4k + \frac{275}{2} - k.\frac{25}{4} - \frac{121}{2} - 50 = 0$$

$$\Rightarrow 4k - \frac{25k}{4} = -\frac{275}{2} + \frac{121}{2} + 50$$

$$\Rightarrow \frac{16k - 25k}{4} = \frac{-275 + 121 + 100}{2}$$

$$\Rightarrow -\frac{9k}{4} = \frac{-54}{2}$$

$$k = 12$$

20. **Prove that the equation $6x^2 + 5xy - 4y^2 + 7x + 13y - 3 = 0$ represent two straight lines; find also their equations, point of intersection and angle between them.**

Solution:

General equation of second degree is

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$$

Given equation is

$$6x^2 + 5xy - 4y^2 + 7x + 13y - 3 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$$

Comparing equation (i) and (ii), we get

$$a = 6, h = \tfrac{5}{2}, b = -4, g = \tfrac{7}{2}, f = \tfrac{13}{2}, c = -3$$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$$\Delta = 6.(-4).(-3) + 2.\frac{13}{2}.\frac{7}{2}.\frac{5}{2} - 6.\left(\frac{13}{2}\right)^2 + 4.\left(\frac{7}{2}\right)^2 + 3\left(\frac{5}{2}\right)^2$$

$$\Delta = 72 + \frac{910}{8} - \frac{1014}{4} + \frac{196}{4} + \frac{75}{4}$$

$$\Delta = \frac{576 + 910 - 2028 + 392 + 150}{8}$$

$$\Delta = \frac{2028 - 2028}{8}$$

$\Delta = 0$

So this given equation can be represents as a pair of straight lines.

Now from (ii) we get

$$6x^2 + x(5y+7) - (4y^2 - 13y + 3) = 0$$

$$\therefore x = \frac{-(5y+7) \pm \sqrt{(5y+7)^2 + 4.6(4y^2 - 13y + 3)}}{2.6}$$

$$= \frac{-(5y+7) \pm \sqrt{25y^2 + 70y + 49 + 96y^2 - 312y + 72}}{12}$$

$$= \frac{-(5y+7) \pm \sqrt{121y^2 - 242y + 121}}{12}$$

$$= \frac{-(5y+7) \pm \sqrt{(11y-11)^2}}{12}$$

$$= \frac{-(5y+7) \pm (11y-11)}{12}$$

$$\Rightarrow 12x + 5y + 7 = \pm(11y - 11)$$

Now for positive sign, we get

$$12x + 5y + 7 = 11y - 11$$

$$\Rightarrow 12x + 5y + 7 - 11y + 11 = 0$$

$$\Rightarrow 12x - 6y + 18 = 0$$

$$\therefore 2x - y + 3 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots (iii)$$

and for negative sign, we get

$$12x + 5y + 7 = -11y + 11$$

$$\Rightarrow 12x + 5y + 7 + 11y - 11 = 0$$

$$\Rightarrow 12x + 16y - 4 = 0$$

$$\therefore 3x + 4y - 1 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots (iv)$$

So, the equation (ii) represent two straight lines $2x - y + 3 = 0$ and $3x + 4y - 1 = 0$.

Point of intersection:

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Solving (iii) and (iv), we get

$x = -1$ and $y = 1$

Intersecting point $(-1, 1)$

Angle:

$$\theta = tan^{-1}\frac{2\sqrt{h^2 - ab}}{a + b}$$

$$= tan^{-1}\frac{2\sqrt{\left(\frac{5}{2}\right)^2 - 6(-4)}}{6 - 4}$$

$$= tan^{-1}\frac{2\sqrt{\frac{25}{4} + 24}}{2}$$

$$= tan^{-1}\sqrt{\frac{25}{4} + 24}$$

$$= tan^{-1}\sqrt{\frac{121}{4}}$$

$$\theta = tan^{-1}\frac{11}{2}$$

21. **Prove that the equation $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$ represent two parallel straight lines and find the distance between them.**

Solution:

$x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0 \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (i)$

$\Rightarrow x^2 + 2(3y + 2)x + (9y^2 + 12y - 5) = 0$

$$\Rightarrow x = \frac{-2(3y + 2) \pm \sqrt{4(3y + 2)^2 - 4.1(9y^2 + 12y - 5)}}{2.1}$$

$$= \frac{-2(3y + 2) \pm 2\sqrt{9y^2 + 12y + 4 - 9y^2 - 12y + 5}}{2}$$

$$= -(3y + 2) \pm \sqrt{9}$$

$\therefore x = -3y - 2 \pm 3$

Now for positive sign, we get

$$x = -3y - 2 + 3$$

$$\therefore x + 3y - 1 = 0$$

and for negative sign, we get

$$x = -3y - 2 - 3$$

$$\therefore x + 3y + 5 = 0$$

So the equation (i) represent two parallel straight lines

$x + 3y - 1 = 0$ … … … … … … … … … (ii)

and $x + 3y + 5 = 0$ … … … … … … … … … (iii)

Now from (ii). We get

$$3y = -x + 1$$

$$\therefore y = -\frac{1}{3}x + \frac{1}{3}$$

Slope of (ii) is $m_1 = -\frac{1}{3}$

and from (iii). We get

$$3y = -x - 5$$

$$\therefore y = -\frac{1}{3}x - \frac{5}{3}$$

Slope of (iii) is $m_2 = -\frac{1}{3}$

$$\because \ m_1 = m_2$$

So the equation (ii) and (iii) are parallel straight lines.

Now the distance between (ii) and (iii), we get

$$= \left|\frac{5 - (-1)}{\sqrt{1^2 + 3^2}}\right|$$

$$= \left|\frac{6}{\sqrt{1 + 9}}\right|$$

$$= \frac{6}{\sqrt{10}}$$

$$= \frac{6\sqrt{10}}{10}$$

$$= \frac{3\sqrt{10}}{5}\ unit$$

22. **Show that $x^2 + 4xy - 2y^2 + 6x - 12y - 15 = 0$ represents pair of straight lines.**
23. **Find the value of k so that the equation $2x^2 + xy - y^2 - 2x - 5y + k = 0$ may represent pair of straight lines.**
24. **Find the value of k so that the equation $6x^2 + xy + ky^2 - 11x + 43y - 35 = 0$ may represent pair of straight lines.**
25. **Find the value of k so that the equation $x^2 - kxy + 2y^2 + 3x - 5y + 2 = 0$ may represent pair of straight lines.**
26. **Prove that the equation $12x^2 - 10xy + 2y^2 + 11x - 5y + 2 = 0$ represent two straight lines; find also their equations, point of intersection and angle between them.**

27. **Prove that the equation $12x^2 - 10xy + 2y^2 + 14x - 5y + 2 = 0$ represent two straight lines; find also their equations, point of intersection and angle between them.**

25. **Prove that the equation $6x^2 - 5xy - 6y^2 + 14x + 5y + 4 = 0$ represent two straight lines; find also their point of intersection and angle between them.**

28. **Prove that the equation $3y^2 - 8xy - 3x^2 - 29x + 3y - 18 = 0$ represent two straight lines; find also their point of intersection and angle between them.**

29. **Prove that the equation $4x^2 + 12xy + 9y^2 - 6x - 9y + 2 = 0$ represent two parallel straight lines and find the distance between them.**